Olist Brazilian E-Commerce Analytics

Client Business Insights & Recommendations

A concise management view of commercial performance, customer behavior, marketplace structure and delivery operations.

99,441	97.02%	€13.59M	3.0%	6.77%	41.10%
Total orders	Delivery completion rate	Product revenue, all statuses	Repeat-customer share in SQL analysis	Late-delivery rate, eligible delivered orders	Revenue from top 10% of sellers

Source basis: Olist Brazilian E-Commerce Public Dataset and the accompanying analytical study. Findings describe the historical dataset analyzed and should not be interpreted as current Olist operating performance.

Prepared from the full analytical study

1. Executive Snapshot

The analysis shows a marketplace with strong recorded order completion and growing historical revenue, alongside three areas that warrant closer management attention: low repeat purchasing, concentrated seller revenue, and delivery variability. The findings support a recurring KPI process that connects commercial, customer and operational indicators.

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What the data says

• **Commercial performance:** 99,441 orders were recorded; 96,478 were delivered, giving a 97.02% completion rate. Product revenue across all statuses was approximately €13.59M.

• **Customer retention:** repeat customers represented about 3% of customers in the SQL analysis, while their average order value was approximately €260.05 versus €137.96 for one-time customers.

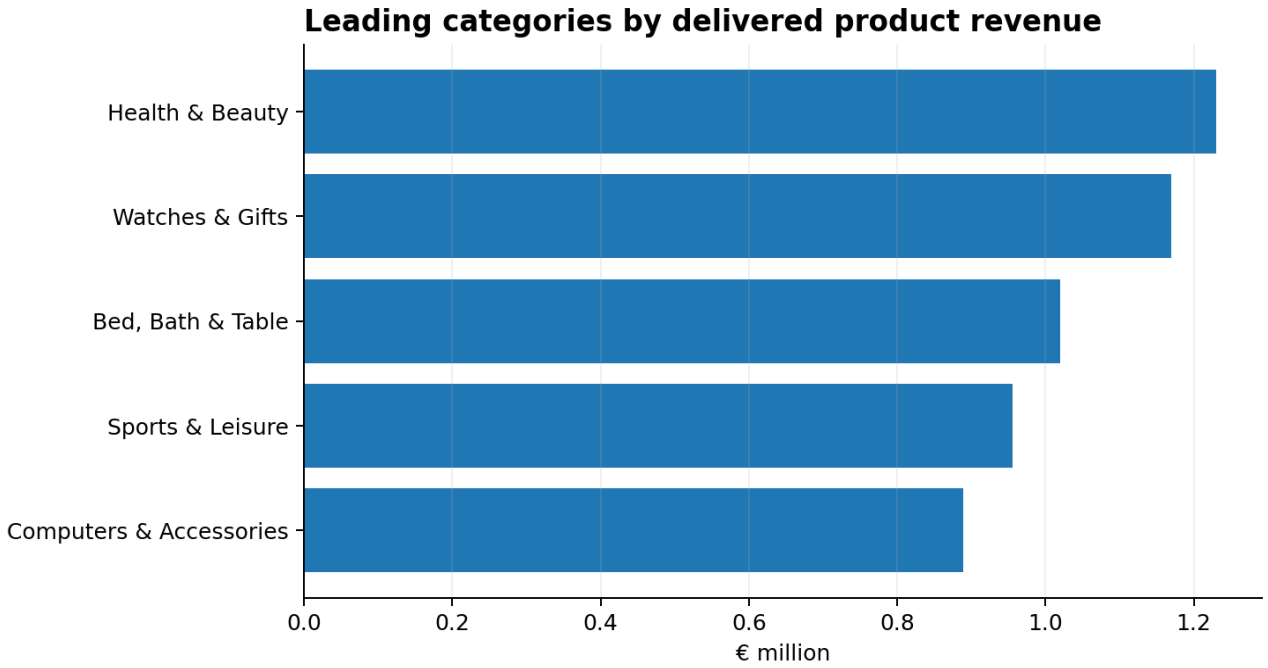
• **Delivery:** 6,535 of 96,476 eligible delivered orders were late, corresponding to a 6.77% late-delivery rate.

• **Seller structure:** the top 10% of sellers accounted for 41.10% of seller revenue and the top 50% for 83.31%.

• **Risk analytics:** the Random Forest model achieved ROC-AUC of about 0.774. At the illustrative 0.50 threshold, it captured about 62.9% of late orders with 18.0% precision; this is a prioritization prototype, not a deployment-ready predictor.

2. Commercial Performance

Historical product revenue increased over the observed period and reached its highest reported monthly level in November 2017 at approximately €1.0M. The analysis does not establish a stable recurring seasonal pattern.

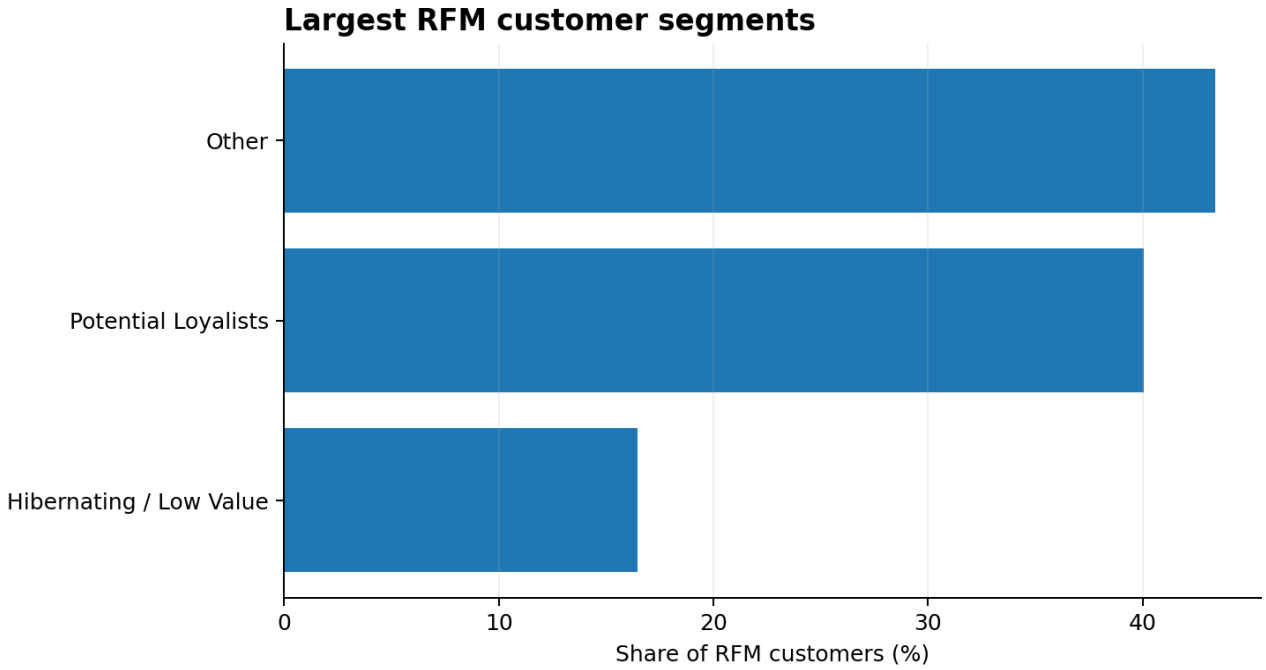


The leading delivered-revenue categories were Health & Beauty (€1.23M), Watches & Gifts (€1.17M), Bed, Bath & Table (€1.02M), Sports & Leisure (€0.955M), and Computers & Accessories (€0.889M). Bed, Bath & Table also recorded the highest unit volume among the leading categories, at approximately 10,953 units.

Business implication: category decisions should combine revenue, units sold and average product price. Revenue leaders and volume leaders are not necessarily the same, so assortment and pricing decisions should use both dimensions.

3. Customer Value & Retention

Customer behavior is characterized by a very high share of one-time purchasing. In the SQL analysis, approximately 97% of customers were one-time purchasers and 3% were repeat purchasers. Repeat customers had a substantially higher average order value.

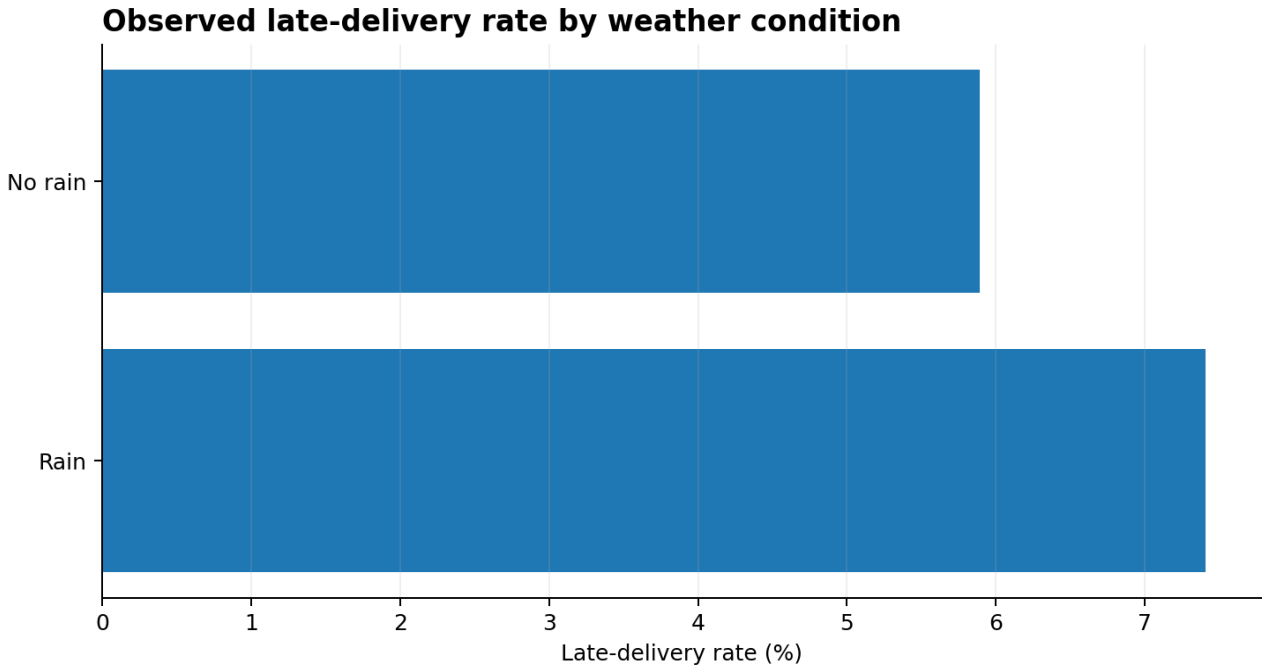


The RFM analysis shows that Potential Loyalists represent 40.06% of the RFM population and contribute 40.95% of analyzed segment revenue. The broad Other segment represents 43.36% of customers and 53.09% of revenue. Hibernating / Low Value customers represent 16.47% of customers but only 5.56% of revenue.

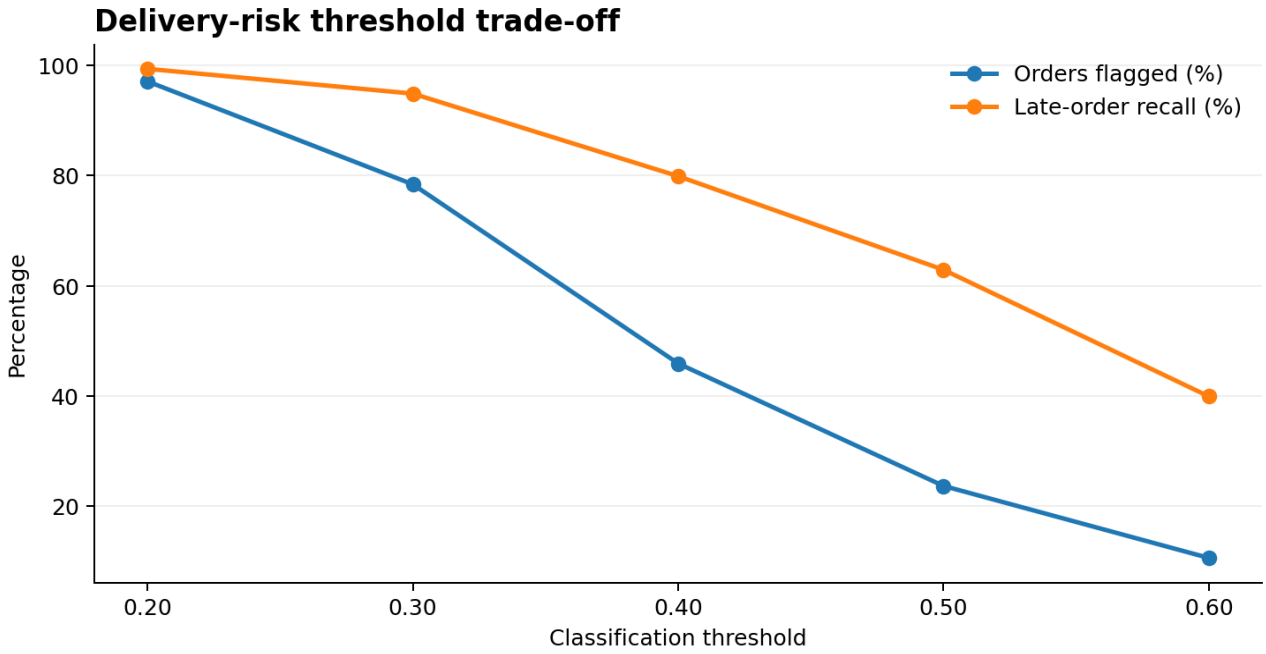
Business implication: retention activity should be differentiated by customer behavior. Potential Loyalists provide a large group for investigating second-purchase conversion; high-value groups require value protection; inactive/low-value customers should be monitored for changes in recency and subsequent purchasing.

4. Delivery Performance & Risk

Overall delivery performance is strong in terms of completion, but 6.77% of eligible delivered orders were late. Delivery performance varies by month and geography, meaning aggregate KPIs can conceal localized issues.



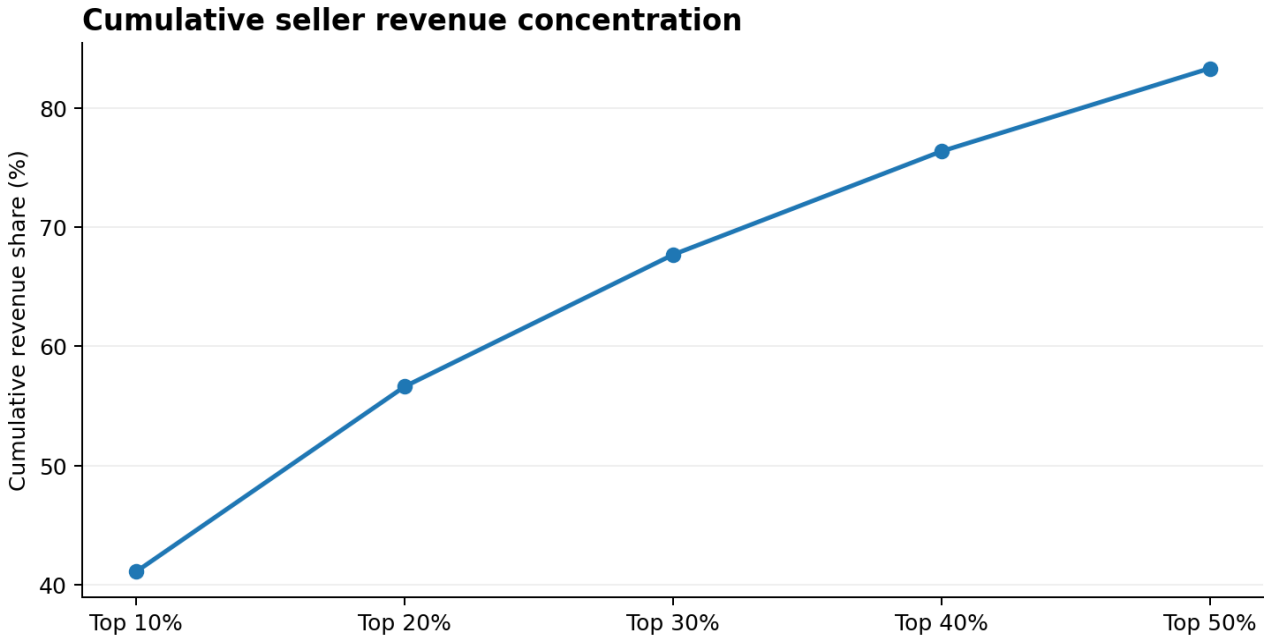
In the 40,621-order weather-matched sample, the late-delivery rate was 5.89% on dates without rain and 7.41% on rainy dates. This is an observed association, not evidence that rainfall caused delays.



The Random Forest model provides a potential screening mechanism. At the illustrative 0.50 threshold, 23.64% of orders were flagged and 62.9% of actual late orders were identified, with 18.0% precision. The threshold should be calibrated against operational review capacity and the consequences of missed versus unnecessary alerts.

Business implication: monitor delivery by month and geography, use risk scores as review-prioritization signals, and validate the model prospectively before operational adoption.

5. Seller Concentration & Marketplace Structure



Seller revenue is materially concentrated. The top 10% of sellers account for 41.10% of revenue, the top 20% for 56.62%, and the top 50% for 83.31%. Revenue concentration alone does not establish operational dependency, because substitutability and seller disruption effects were not measured.

Business implication: monitor seller revenue contribution together with order volume and delivery performance. Track concentration over time to identify whether marketplace revenue is becoming increasingly dependent on a smaller seller group.

High-revenue seller locations in the analysis included Guariba (São Paulo), Lauro de Freitas (Bahia), Ibitinga (São Paulo), Sumaré (São Paulo) and Itaquaquecetuba (São Paulo). These figures describe seller activity associated with those locations and should not be interpreted as city-wide market performance.

6. Business Recommendations

1. Strengthen customer retention

Monitor repeat-customer rate, average order value and RFM segments. Focus retention analysis on Potential Loyalists, while differentiating treatment for high-value and inactive customers. Test second-purchase and personalized engagement approaches rather than applying broad discounts indiscriminately.

2. Establish delivery-risk monitoring

Use delivery-risk scores as a prioritization signal for operational review. The Random Forest prototype captured about 62.9% of late orders at the illustrative 0.50 threshold, but precision was only about 18%; prospective validation is required before operational deployment.

3. Monitor delivery by geography and time

Track late-delivery rate, average delivery duration and order volume by month and customer geography. Investigate sustained deterioration rather than isolated high rates, particularly where order volumes are small.

4. Manage seller concentration

Track seller revenue contribution, order volume and delivery performance alongside cumulative concentration. Monitor whether marketplace revenue becomes increasingly concentrated among a smaller group of sellers.

5. Use category-level performance for product decisions

Evaluate category revenue, units sold and average price together. Use this view to distinguish high-volume categories from higher-value categories and to guide assortment, pricing and cross-selling investigations.

6. Establish a recurring KPI monitoring framework

Use Power BI as the monitoring layer, supported by the SQL/Python analytical workflow. Review commercial, customer, delivery, seller and risk indicators together rather than in isolation.

7. Management KPI Framework

The Power BI dashboard can serve as the recurring reporting layer, with SQL and Python workflows providing the analytical foundation. KPIs should be reviewed together so that changes in revenue, retention, delivery reliability and seller concentration are interpreted in context.

Business area	Core KPI	Management use
Sales	Orders; product revenue; AOV	Track marketplace evolution and transaction value
Customers	Repeat rate; average spend; RFM mix	Monitor retention and customer value
Delivery	Late rate; average delivery days	Detect service deterioration
Delivery risk	Risk score; flagged share; recall	Prioritize operational review
Sellers	Revenue contribution; concentration	Monitor seller dependency
Products	Category revenue; units; average price	Evaluate category dynamics

Suggested monitoring cycle

1. Review customer retention and RFM composition regularly.
2. Monitor delivery rate and duration by month and geography.
3. Track seller contribution and concentration alongside operational performance.
4. Review category revenue, units and average price together.
5. Re-evaluate predictive-model performance on genuinely future orders before operational use.

8. Client Takeaways

The analysis provides a practical foundation for ongoing e-commerce performance monitoring. The main commercial opportunity identified is to understand and develop repeat purchasing, while the main operational focus is to monitor delivery reliability at a more granular level. Seller concentration and category-level performance should be tracked alongside these indicators to avoid relying on aggregate revenue alone.

The Power BI dashboard consolidates the core views into Sales Overview, Product & Category Intelligence, Orders & Delivery, Customer Overview and Seller Overview, providing a single interface for recurring analysis.

Scope & interpretation

The analysis is based on a historical Olist dataset. It does not provide a complete profitability assessment because comprehensive cost data are unavailable. Weather findings are associative rather than causal, and the late-delivery model requires temporal validation and further operational testing before deployment. KPI values also use different analytical populations, so each metric should be interpreted with its defined denominator and scope.

Source: Olist Brazilian E-Commerce Analytics — End-to-End Business Intelligence and Predictive Analytics Study.